

A Linear Positivity Region for Thue Morse Pressure

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Abstract

We prove a positive region of linear width beyond the first nonlinear feedback boundary in integer Thue–Morse pressure. The response and pressure coefficients at offset s are positive whenever $m \geq 2048$ and $128s + 64 \leq m$. The proof improves both parts of the preceding estimate: complete multipartite matching counts strengthen the marked-pair denominator, a telescoping trigonometric inequality gives exponential bounds for all auxiliary moments, and a weighted resolvent estimate controls the restored insertion and feedback on a fixed complex disk. The full proposed positive range below degree $6m$ remains open.

Status. This is unrefereed ordinary mathematics with exact arithmetic checks, not a Lean formalization. It is a separate strengthening of the preceding report; earlier artifacts are unchanged. No priority or optimal-constant claim is made.

1 The linear region

For $m \geq 2$ put $d = 2m$ and consider

$$(V_a f)(x) = \sum_{2y=x \bmod 1} (\cos \pi y + a \sin \pi y)^d f(y).$$

Let $h_a(0) = 1$ normalize the local analytic eigenfunction, with eigenvalue $\lambda_m(a)$. Define

$$H_{m,r} = [a^{2r}]h_a(1/2), \quad E_{m,r} = [t^{2m+2r}]p_m(t/\pi).$$

Write $R_j = \sum_{w \in \mathbb{Z}+1/2} (\pi w)^{-2j}$, so $H_{m,0} = R_m$ and $0 < R_j \leq R_1 = 1$. The exact feedback identities are

$$\lambda_m(a) = 1 + a^d h_a(1/2), \quad p_m(t/\pi) = d \log \cos t + \log \lambda_m(\tan t). \quad (1)$$

The preceding triangle and boundary results give $H_{m,r}, E_{m,r} > 0$ for $1 \leq r \leq m$ [2, 3]. A companion report establishes fixed-offset asymptotics and a logarithmic-width sufficient condition [4]. Here the offset may be a fixed positive fraction of m .

Theorem 1.1. *For integers $s \geq 0$ and $m \geq 2048$,*

$$128s + 64 \leq m \implies H_{m,m+s} > 0, \quad E_{m,m+s} > 0. \quad (2)$$

Equivalently, $d \geq 4096$ and $128(2s + 1) \leq d$ suffice.

Together with the earlier triangle, this proves positivity for all integer indices

$$1 \leq r \leq m + \left\lfloor \frac{m - 64}{128} \right\rfloor, \quad m \geq 2048.$$

The pressure degree at the upper end is $4m + 2\lfloor(m - 64)/128\rfloor$. This remains far below the conjectural full range $2m < \deg < 6m$. Small- m response positivity cannot extend without qualification: $H_{2,3} = -488/27$ [3].

2 Stronger marked pairs and tangent moments

For a summable nonnegative sequence x , write

$$B_x(a) = \prod_i (1 + ax_i), \quad \mathcal{C}_{d,q}(x) = [a^q] \prod_i ((1 + ax_i)^d - (ax_i)^d).$$

The latter polynomial keeps only monomials whose exponent at each coordinate is at most $d - 1$.

Lemma 2.1. *Let $q = d + 2s$, $k = 2s + 1$, and $d \geq 2s + 2$. Every color profile of q labeled units with multiplicities at most $d - 1$ has at least*

$$(d - 1)_k = (d - 1)(d - 2) \cdots (d - k)$$

matchings of k disjoint pairs using different colors. Consequently, for $a > 0$,

$$\mathcal{C}_{d,q}(x) \leq \frac{d^{2k}}{k!(d - 1)_k} a^{2k-q} e_2(x)^k B_x(a)^d. \quad (3)$$

Proof. Let $M_k(\alpha)$ count the matchings for a color vector α . We prove by induction on k that moving one unit from a smaller class to a larger class cannot increase M_k . The case $k = 0$ is constant. For a move $(a, b) \mapsto (a + 1, b - 1)$ with $a \geq b \geq 1$, remove the moved unit. Matchings that leave it unmatched, or pair it with another color, cancel in the difference. Thus

$$\begin{aligned} M_k(a + 1, b - 1, \gamma) - M_k(a, b, \gamma) \\ = (b - 1)M_{k-1}(a, b - 2, \gamma) - aM_{k-1}(a - 1, b - 1, \gamma). \end{aligned}$$

If $b = 1$ the first term is absent. Otherwise the induction hypothesis makes its M value at most the second one. The difference is nonpositive.

Fill the largest class to $d - 1$, then merge the remaining units into one class. The remaining total is $k \leq d - 1$, so all moves respect the cap. The minimizing profile is therefore $(d - 1, k)$, whose k -matchings number exactly $(d - 1)_k$.

For finitely many coordinates introduce independent variables y_i and

$$\mathcal{D}_x = \sum_{i < j} x_i x_j \partial_{y_i} \partial_{y_j}.$$

In $a^{2k} \mathcal{D}_x^k \prod_i (1 + y_i)^d / k!$, evaluated at $y_i = ax_i$, the coefficient of each original monomial is multiplied by its number of unordered labeled-unit matchings. The derivatives choose distinct units, and their $k!$ orders cancel the denominator. At positive a , each derivative costs at most $dx_i / (1 + ax_i)$. Drop these denominators, sum the pairs, and divide by $a^q (d - 1)_k$ to obtain (3). Monotone convergence gives the infinite-coordinate version. $\square$

Set $\theta_j = \pi / 2^{j+1}$ and, for positive odd n , $x_j(n) = |\tan(n\theta_j)| / n$. Define

$$S_s = \sum_{\substack{n \geq 3 \\ n \text{ odd}}} n^{2s} e_2(x(n))^{2s+1}.$$

Lemma 2.2. *For all $s \geq 0$,*

$$S_s \leq (4\pi)^{2s} S_0 < 11(4\pi)^{2s}. \quad (4)$$

Proof. For $0 \leq a \leq 1/2$ and real θ ,

$$(|\cos \theta| + a|\sin \theta|)(1 + |\sin \theta|) \leq 1 + |\sin 2\theta|. \quad (5)$$

Put $u = |\sin \theta|$, $c = |\cos \theta|$. If $u > 0$, the inequality is equivalent to

$$a(1+u) \leq c + \frac{u}{1+c}.$$

The left side is at most one, and the right side is at least $c + u^2/(1+c) = 1$. The case $u = 0$ is equality.

Multiply (5) along $n\theta_j$, $j = 1, \dots, N$. Since $|\sin(n\pi/2)| = 1$, the product is at most two. Its infinite limit exists locally uniformly in a , with nonnegative coefficients. At $a = 1/2$ its degree-two coefficient is at most eight. The dyadic sine-product identity gives

$$\prod_{j \geq 1} |\cos(n\theta_j)| = \frac{2}{\pi n},$$

so that coefficient is $2e_2(|\tan(n\theta_j)|)/(\pi n)$. Therefore $ne_2(x(n)) \leq 4\pi$, uniformly in odd n . Multiplying this inequality to power $2s$ by $e_2(x(n))$ and summing proves the first part of (4). The earlier two-insertion estimate gives $S_0 < 11$ [3]. $\square$

The same argument also yields $e_j(|\tan(n\theta_\ell)| : \ell \geq 1)/n \leq \pi 2^j$ for every fixed j .

3 A fixed disk for the capped resolvent

Let $A = V_0$, $h_0 = h_{a=0}$, $Pf = f(0)h_0$, $Q = I - P$, and $\text{ev } f = f(1/2)$. Work in the invariant Fourier space with indices $1-m, \dots, m-1$. Its atomic eigenvalues at zero are 2^{-j} , $0 \leq j \leq 2m-2$, so the eigenvalue one is simple [1, 2].

Write

$$V_a = \sum_{j=0}^d a^j V_j, \quad V_{<}(a) = \sum_{j=0}^{d-1} a^j V_j.$$

Evaluation at zero is preserved by $V_{<}$. Let $f_{<}(a)$ be its normalized fixed vector, and put

$$T_{<}(a) = (I - QV_{<}(a)Q)^{-1} \text{ on } Q, \quad Z(a) = \text{ev } f_{<}(a).$$

For $x \in [0, 1]$ define, periodically,

$$g(x) = \sin \pi x, \quad H(x) = 1 + \frac{1}{8} \sin \pi x, \quad K_d(x) = H(x)^d g(x).$$

Use $\|f\|_K = \sup |f|/K_d$ on vectors vanishing at zero.

Lemma 3.1. *For $d \geq 128$ and complex $|a| \leq 1/16$, the restriction of $V_{<}(a)$ to Q has K -norm at most $3/4$. Hence $\|T_{<}(a)\|_K \leq 4$ throughout this disk.*

Proof. Put $a_0 = 1/16$, $\beta = 1/8$, $u = \sin(\pi x/2)$, $c = \cos(\pi x/2)$, and $N = 1 + 2\beta uc$. Define

$$p_1 = \frac{(c + a_0 u)(1 + \beta u)}{N}, \quad r_1 = \frac{a_0 u(1 + \beta u)}{N},$$

$$p_2 = \frac{(u + a_0 c)(1 + \beta c)}{N}, \quad r_2 = \frac{a_0 c(1 + \beta c)}{N}.$$

The absolute capped operator applied to K_d , divided by $K_d(x)$, is

$$\frac{p_1^d - r_1^d}{2c} + \frac{p_2^d - r_2^d}{2u}, \tag{6}$$

with endpoint limits. This majorizes complex a because the sum of absolute coefficients at radius a_0 is $(|\cos \pi y| + a_0 \sin \pi y)^d - (a_0 \sin \pi y)^d$.

Both $p_i \leq 1$. Indeed,

$$N - (c + a_0 u)(1 + 2a_0 u) = \frac{u^2}{1+c} + a_0 u(2c-1) - 2a_0^2 u^2.$$

If $c \geq 1/2$, this is at least $u^2(1/2 - 2a_0^2)$. If $c \leq 1/2$, it is at least $1/2 - a_0 - 2a_0^2 = 55/128$.

When $u \leq 1/4$, the first term of (6) is at most $2/\sqrt{15} < 8/15$. The second is at most

$$\frac{9d}{16}(45/128)^{d-1},$$

since $p_2 \leq 45/128$ and $p_2 - r_2 = u(1 + \beta c)/N$. For $d \geq 4$ this decreases with d , and at $d = 4$ its sum with $8/15$ is below $3/4$. The case $c \leq 1/4$ is symmetric.

In the remaining interior, each $p_i \leq 39/40$. If $c \geq 1/2$, its normalized numerator gap is at least $(63/2048)/(9/8) = 7/256 > 1/40$; if $c \leq 1/2$ it is even larger. Thus (6) is at most $4(39/40)^d \leq 4(39/40)^{128} < 3/4$. The Neumann series on Q completes the proof. $\square$

Put

$$L = \frac{153}{128}, \quad M = \frac{1377}{1024}, \quad J = L^2 = \frac{23409}{16384}.$$

The elementary insertion bounds, with $1 \leq j < d$, are

$$Ag \leq g/2, \quad |V_j f| \leq \frac{1}{2} \binom{d}{j} \|f\|_\infty g, \quad \|h_0\|_\infty \leq 1.$$

They follow from the two branches and weighted AM–GM as in [3, 4]. Since $K_d \geq g$, Lemma 3.1 gives

$$\|f_{<} - h_0\|_K \leq 2(17/16)^d, \quad \|f_{<}\|_\infty \leq 3L^d.$$

The initial h_0 is used in sup norm; it does not vanish at zero.

For a trigonometric polynomial v of degree at most $m-1$, Fourier coefficient bounds give $\|v'\|_\infty \leq 2\pi m(m-1)\|v\|_\infty$. Since $QV_d f$ vanishes at zero and $\|QV_d f\|_\infty \leq 2\|f\|_\infty$,

$$\|QV_d f\|_K \leq \|QV_d f\|_g < 2d^2\|f\|_\infty.$$

Therefore, throughout $|a| \leq 1/16$,

$$|\text{ev } T_{<} QV_d f_{<}| \leq 24d^2 M^d, \quad |Z \text{ ev } T_{<} (f_{<} - h_0)| \leq 24J^d. \quad (7)$$

For the second bound, $\|T_{<} (f_{<} - h_0)\|_\infty \leq 8L^d$.

4 The exact response comparison

The finite-iterate periodization yields the absolutely convergent capped orbit formula

$$[a^q]Z(a) = \sum_{w \in \mathbb{Z} + 1/2} (\pi w)^{-d} [a^q] \prod_{j \geq 1} ((1 + a \tan(\pi w/2^j))^d - (a \tan(\pi w/2^j))^d). \quad (8)$$

At every fixed coefficient all insertion degrees are below d . Their absolute values map bounded functions into multiples of g , and all zero-degree gaps contract g by $1/2$. Summing over the finitely many degree compositions proves absolute convergence and the passage from finite iterates [4].

Let $n = 2s < d$ and $q = d + n$. Applying Q to the true eigenvalue equation and subtracting the capped fixed-vector equation gives

$$h_a - f_{<} = a^d T_{<} (QV_d h_a - h_a(1/2)(h_a - h_0)).$$

At total order $d + n < 2d$, replace h_a in the braces by $f_{<}$. Thus

$$H_{m,m+s} = [a^q]Z + [a^n](\text{ev } T_{<} QV_d f_{<} - Z \text{ ev } T_{<}(f_{<} - h_0)). \quad (9)$$

Cauchy's inequality and $M < J$ bound the second term in absolute value by

$$24(d^2 + 1)J^d 16^n. \quad (10)$$

This is the key improvement over low-order polynomial-in- d jet bounds. No sign of the frozen response is assumed.

5 Three exponentially small errors

Let $b_j = \tan(\pi/2^{j+1})$ and $B(a) = \prod_j (1 + ab_j)$. The strict tangent comparison from [3] is

$$B_{x(n)}(a) \leq \eta(a)B(a), \quad \eta(a) = \frac{1 + \theta a}{1 + a}, \quad \theta = \frac{1 + \sqrt{2}}{3} < \frac{81}{100}.$$

It follows from weak log-majorization by $(\theta, b_2, b_3, \dots)$; the earlier report supplies the dyadic single-crossing proof.

Assume $d \geq 4096$ and $k = 2s + 1 \leq d/128$. Choose a_q with $a_q B'(a_q)/B(a_q) = q/d$. The exact half-angle certificate gives $a_q > 4/5$ [3]. For the upper bound, $q/d < 33/32$ and

$$B'(1)/B(1) > \frac{1}{2} + \frac{3}{11} + \frac{3}{19} + \frac{3}{35} + \frac{3}{67} > 33/32,$$

using $\tan x > x$ and $\pi > 3$. Thus $a_q < 1$. Consequently

$$\eta(a_q) < 206/225, \quad B(a_q)/a_q^{q/d} > 3.$$

Indeed, using $b_1 = 1, b_2 > 2/5, b_3 > 1/6$,

$$\frac{B(a_q)}{a_q^{q/d}} \geq \frac{B(a_q)}{a_q} > \frac{1}{a_q} + \frac{47}{30} + \frac{19a_q}{30} + \frac{a_q^2}{15} > 3.$$

The a_q -tilted countable Bernoulli sum has integer mean q and variance at most q . The mean is a mode by [5, Theorem 2]; Chebyshev's inequality then gives

$$E_0 := 2(2/\pi)^d [u^q]B(u)^d > \frac{(21/11)^d}{4\sqrt{q}}. \quad (11)$$

Here E_0 is the unmodified nearest-orbit contribution. We used $[u^q]B(u)^d \geq B(a_q)^d a_q^{-q}/(8\sqrt{q})$ and $\pi < 22/7$.

Remote orbits. Lemmas 2.1 and 2.2 bound their absolute sum divided by E_0 by

$$8\sqrt{q} \frac{d^{2k} S_s}{k!(d-1)_k} \eta(a_q)^d.$$

Use $S_s < 11 \cdot 13^k$, $(d-1)_k \geq (127d/128)^k$, and $k! \geq (k/3)^k$. For $v = k/d \leq 1/128$, the increasing function

$$v \log \frac{39 \cdot 128}{127v}$$

is at most $\log(39 \cdot 128^2/127)/128 < 9/128$. Hence the remote ratio is at most

$$132\sqrt{d} \exp((9/128 - 19/225)d).$$

Its logarithm divided by d is bounded by

$$\frac{19}{8192} + \frac{9}{128} - \frac{19}{225} < -\frac{1}{100}.$$

Here and below we use $d \geq 4096$, $\log 4096 < 9$, and the decreasing function $(c + b \log d)/d$.

Deleted nearest monomials. Since $q < 2d$, deleting full powers from the nearest product removes exactly

$$\sum_j b_j^d [u^n] \prod_{i \neq j} (1 + ub_i)^d \leq \frac{2(2d)^n}{n!}.$$

The relative cost is at most $16\sqrt{q}(2d)^n/(n!3^d)$. For $n > 0$, use $n! \geq (n/3)^n$ and $n/d \leq 1/128$; for $n = 0$ the numerator factor is one. The normalized logarithm is at most

$$\frac{17}{8192} + \frac{8}{128} - 1 < -\frac{1}{2}.$$

Restoration and feedback. Divide (10) by (11). The rational inequality

$$J \frac{11}{21} = \frac{257499}{344064} < \frac{3}{4}$$

gives the relative bound $288d^{5/2}(3/4)^d 16^n$. Its normalized logarithm is at most

$$\frac{57}{8192} - \frac{29}{128} < -\frac{1}{5},$$

using $\log 16 < 3$ and $\log(4/3) > 1/4$.

Each of these three errors is less than $E_0/8$. Thus

$$H_{m,m+s} > \frac{E_0}{2} > \frac{(21/11)^d}{8\sqrt{q}} > 0. \quad (12)$$

The factorial bound follows from $\log(k!) \geq \int_1^k \log x \, dx \geq k \log k - k$ and $e < 3$. All displayed rational margins and elementary exponential bounds are checked exactly in the source bundle.

6 Pressure transfer and scope

Every smaller offset satisfies the same linear condition. The pre-feedback triangle and (12) make all response terms in

$$\begin{aligned} E_{m,m+s} &= \sum_{j=0}^{m+s} H_{m,j} [t^{2(m+s-j)}] (\tan t/t)^{d+2j} \\ &\quad - \frac{1}{2} [t^n] (\tan t/t)^{2d} h_{\tan t} (1/2)^2 - \frac{m}{2m+s} R_{2m+s} \end{aligned}$$

positive, apart from the two explicit subtractions. This formula follows from (1); the cubic logarithm begins at degree $3d > 2d + n$.

Take $|t| = \rho = 1/32$. Positive tangent coefficients give

$$|\tan t| \leq \tan \rho \leq \rho/(1 - \rho^2) < 1/16.$$

The coefficients through degree $n < d$ of the true response equal those of $f_{<}$. Cauchy's inequality and (7) bound the quadratic subtraction by

$$\frac{9}{2} [J(1024/1023)^2]^d 32^n.$$

The cosine subtraction is at most one, so their sum is bounded by the same expression with $9/2$ replaced by $11/2$. The exact inequality

$$J(1024/1023)^2 \frac{11}{21} < \frac{3}{4}$$

then bounds its ratio to (12) by $66\sqrt{d}(3/4)^d 32^n$. Its normalized logarithm is at most

$$\frac{19}{8192} - \frac{113}{512} < 0,$$

using $\log 32 < 15/4$. The single response term $H_{m,m+s}$ therefore exceeds both subtractions, completing Theorem 1.1.

The bundle includes exact checks of the strengthened matching multiplicity on 2,632 finite profiles, all branch-contraction and exponential-margin constants, and the earlier response identity checks. These supplement the universal analytic proof; no negative finite search is used as a proof.

The remaining pressure interval up to degree $6m$, the optimal linear-width constant, and the first later negative pressure degree remain open. The fixed-saddle asymptotic of [4] is not asserted uniformly in this linear region; a moving saddle would be required for such a statement.

References

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